

Overview: Here we used Langchain/Langgraph frameworks to create a simple AI agents to act as a Fidelity financial advisor to answer user questions about retirement planning and retirement mutual funds.

Requirements:

1. Use Langchain/Langgraph framework to create an IPYNB script to build the graph of an AI agent that will act as a Fidelity financial advisor to answer user questions about retirement planning.
2. Use WebBaseLoader from Langchain to load the following webpage into a document:
 - <https://www.fidelity.com/viewpoints/retirement/how-much-do-i-need-to-retire>
3. Store the document into Chroma vector store
4. Create a retriever_tool that will be used by the AI Agent to verify that the document is relevant to the question asked by the user before sending it to the chain of the generative AI model to generate an answer to the user question.
5. Create a chain to send the following prompt to LLM to determine if the document(context) that will be used to answer the user's question is relevant or not:

```
prompt = PromptTemplate(
    template="""You are a financial advisor answering user question about retirement planning. \n
    Here is the retrieved document: \n\n {context} \n\n
    Here is the user question: {question} \n
    If the document contains keyword(s) or semantic meaning related to the user question, grade it as relevant. \n
    Give a binary score 'yes' or 'no' score to indicate whether the document is relevant to the question.""",
    input_variables=["context", "question"],
)
```

6. Send the following prompt to your graph:

```
inputs = {
    "messages": [
        ("user", "What are the steps that I should take to determine how much I need to save for retirement"),
    ]
}
```

7. Inspect and comment on the output produced by your Ai Agent based on the input prompt you entered above; your output should contain content similar to the following:

```
{ 'messages': [ 'To determine how much you need to save for retirement, start '
                'by aiming to save a multiple of your salary by certain ages: '
                '1x by 30, 3x by 40, 6x by 50, 8x by 60, and 10x by 67. '
                'Consider factors such as your desired retirement age and '
                "lifestyle, as these will impact your savings goal. If you're "
                'behind, remember that there are strategies to catch up, so '
                'take action to improve your savings plan.']]}
```

8. Enter another prompt that has "similar" semantic to the prompt entered above. Inspect and comment on the output generated by the AI Agent.
9. Enter a prompt that is not relevant to retirement planning. Inspect and comment on the output generated by the AI Agent.

	Allocation																																																					
Fidelity Freedom® 2045 Fund (FFFGX)																																																						
No Transaction Fee ⁴																																																						
Hypothetical Growth of \$10,000 ^{6,7} AS OF 11/30/2024 ; Target-Date 2045 ● FFFGX : \$23,898 ● S&P 500 Index : \$35,002 ● Target-Date 2045 : \$22,139																																																						
The performance data featured represents past performance, which is no guarantee of future results. Investment return and principal value of an investment will fluctuate; therefore, you may have a gain or loss when you sell your shares. Current performance may be higher or lower than the performance data quoted.	Morningstar® Snapshot ^{*11} AS OF 11/30/2024 Morningstar Category Target-Date 2045 Risk of this Category Overall Rating Returns Expenses *Data provided by Morningstar																																																					
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Details Morningstar Category Target-Date 2045 Fund Inception 06/01/2006 NAV 12/19/2024 \$13.73 Exp Ratio (Gross) 05/30/2024 0.75% Exp Ratio (Net) 05/30/2024 0.75% Minimum to Invest \$0.00 Turnover Rate 09/30/2024 13.00% Portfolio Net Assets (\$M) 11/30/2024 \$21,466.13 Share Class Net Assets (\$M) 11/30/2024 \$4,597.27																																																						
Equity StyleMap ^{*3} AS OF 10/31/2024 Historical Current Value Blend Growth LG MD SM Large Blend *91.96% Fund Assets Covered																																																						
Fund Manager(s) Co-Manager : Andrew J. Dierdorf (since 06/21/2011) Co-Manager : Brett F. Sumson (since 01/21/2014)																																																						

- "What is the name of this fund?"
- "Who is the fund manager?"
- "What is the calendar year return for 2022 for this fund and S&P 500?"
- "What is the Portfolio Net Assets?"

- "What is the Morningstar rating for this fund? How many funds used to rate this fund?"

Using all the above requirements Develop IPYNB scripts to run successfully the following **8 Experiments**. You MUST create a NEW script and name it properly for every experiment:

Experiment Number	Platform	Model		Framework
1	Ollama [Self-Hosted]	llama3.2:3b [2.0 GB]	HuggingFaceEmbedding BAAI/bge-small-en-v1.5	LangChain/LangGraph
2	Replicate [Cloud]	meta/meta-llama-3-70b-instruct	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph
3	OpenAI [Cloud]	gpt-4o-mini	text-embedding-3-small	LangChain/LangGraph
4	Ollama [Self-Hosted]	llama3.2:3b [2.0 GB]	HuggingFaceEmbedding BAAI/bge-small-en-v1.5	LlamaIndex
5	Replicate [Cloud]	meta/meta-llama-3-70b-instruct	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LlamaIndex
6	OpenAI [Cloud]	gpt-4o-mini	text-embedding-3-small	LlamaIndex
7	Replicate [Cloud]	deepseek-ai/deepseek-r1	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph
8	Replicate [Cloud]	deepseek-ai/deepseek-r1	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LlamaIndex